

Section 1: Principles of chemistry

- a) States of matter
- b) Atoms
- c) Atomic structure
- d) Relative formula masses and molar volumes of gases
- e) Chemical formulae and chemical equations
- f) Ionic compounds
- g) Covalent substances
- h) Metallic crystals
- i) Electrolysis

a) States of matter

Students will be assessed on their ability to:

- 1.1 understand the arrangement, movement and energy of the particles in each of the three states of matter: solid, liquid and gas
- 1.2 describe how the interconversion of solids, liquids and gases are achieved and recall the names used for these interconversions
- 1.3 describe the changes in arrangement, movement and energy of particles during these interconversions.

b) Atoms

Students will be assessed on their ability to:

- 1.4 describe simple experiments leading to the idea of the small size of particles and their movement including:
 - i dilution of coloured solutions
 - ii diffusion experiments
- 1.5 understand the terms atom and molecule
- 1.6 understand the differences between elements, compounds and mixtures
- 1.7 describe techniques for the separation of mixtures, including simple distillation, fractional distillation, filtration, crystallisation and paper chromatography.

c) Atomic structure

Students will be assessed on their ability to:

- 1.8 recall that atoms consist of a central nucleus, composed of protons and neutrons, surrounded by electrons, orbiting in shells
- 1.9 recall the relative mass and relative charge of a proton, neutron and electron
- 1.10 understand the terms atomic number, mass number, isotopes and relative atomic mass (A_r)
- 1.11 calculate the relative atomic mass of an element from the relative abundances of its isotopes
- 1.12 understand that the Periodic Table is an arrangement of elements in order of atomic number
- 1.13 deduce the electronic configurations of the first twenty elements from their positions in the Periodic Table
- 1.14 deduce the number of outer electrons in a main group element from its position in the Periodic Table.

d) Relative formula masses and molar volumes of gases

Students will be assessed on their ability to:

- 1.15 calculate relative formula masses (M_r) from relative atomic masses (A_r)
- 1.16 understand the use of the term mole to represent the amount of substance
- 1.17 understand the term mole as the Avogadro number of particles (atoms, molecules, formulae, ions or electrons) in a substance**
- 1.18 carry out mole calculations using relative atomic mass (A_r) and relative formula mass (M_r)
- 1.19 understand the term molar volume of a gas and use its values (24 dm^3 and $24,000\text{ cm}^3$) at room temperature and pressure (rtp) in calculations.**

e) Chemical formulae and chemical equations

Students will be assessed on their ability to:

- 1.20 write word equations and balanced chemical equations to represent the reactions studied in this specification
- 1.21 use the state symbols (s), (l), (g) and (aq) in chemical equations to represent solids, liquids, gases and aqueous solutions respectively
- 1.22 understand how the formulae of simple compounds can be obtained experimentally, including metal oxides, water and salts containing water of crystallisation
- 1.23 calculate empirical and molecular formulae from experimental data
- 1.24 calculate reacting masses using experimental data and chemical equations
- 1.25 calculate percentage yield**
- 1.26 carry out mole calculations using volumes and molar concentrations.

f) Ionic compounds

Students will be assessed on their ability to:

- 1.27 describe the formation of ions by the gain or loss of electrons
- 1.28 understand oxidation as the loss of electrons and reduction as the gain of electrons
- 1.29 recall the charges of common ions in this specification
- 1.30 deduce the charge of an ion from the electronic configuration of the atom from which the ion is formed
- 1.31 explain, using dot and cross diagrams, the formation of ionic compounds by electron transfer, limited to combinations of elements from Groups 1, 2, 3, and 5, 6, 7
- 1.32 understand ionic bonding as a strong electrostatic attraction between oppositely charged ions
- 1.33 understand that ionic compounds have high melting and boiling points because of strong electrostatic forces between oppositely charged ions
- 1.34 understand the relationship between ionic charge and the melting point and boiling point of an ionic compound**
- 1.35 describe an ionic crystal as a giant three-dimensional lattice structure held together by the attraction between oppositely charged ions**
- 1.36 draw a simple diagram to represent the positions of the ions in a crystal of sodium chloride.**

g) Covalent substances

Students will be assessed on their ability to:

- 1.37 describe the formation of a covalent bond by the sharing of a pair of electrons between two atoms
- 1.38 understand covalent bonding as a strong attraction between the bonding pair of electrons and the nuclei of the atoms involved in the bond
- 1.39 explain, using dot and cross diagrams, the formation of covalent compounds by electron sharing for the following substances:
 - i hydrogen
 - ii chlorine
 - iii hydrogen chloride
 - iv water
 - v methane
 - vi ammonia
 - vii oxygen
 - viii nitrogen
 - ix carbon dioxide
 - x ethane
 - xi ethene

- 1.40 recall that substances with simple molecular structures are gases or liquids, or solids with low melting points
- 1.41 explain why substances with simple molecular structures have low melting points in terms of the relatively weak forces between the molecules
- 1.42 explain the high melting points of substances with giant covalent structures in terms of the breaking of many strong covalent bonds
- 1.43 draw simple diagrams representing the positions of the atoms in diamond and graphite**
- 1.44 explain how the uses of diamond and graphite depend on their structures, limited to graphite as a lubricant and diamond in cutting.**

h) Metallic crystals

Students will be assessed on their ability to:

- 1.45 describe a metal as a giant structure of positive ions surrounded by a sea of delocalised electrons
- 1.46 explain the malleability and electrical conductivity of a metal in terms of its structure and bonding.

i) Electrolysis

Students will be assessed on their ability to:

- 1.47 understand an electric current as a flow of electrons or ions
- 1.48 understand why covalent compounds do not conduct electricity
- 1.49 understand why ionic compounds conduct electricity only when molten or in solution
- 1.50 describe simple experiments to distinguish between electrolytes and non-electrolytes
- 1.51 recall that electrolysis involves the formation of new substances when ionic compounds conduct electricity
- 1.52 describe simple experiments for the electrolysis, using inert electrodes, of molten salts such as lead(II) bromide
- 1.53 describe simple experiments for the electrolysis, using inert electrodes, of aqueous solutions of sodium chloride, copper(II) sulfate and dilute sulfuric acid and predict the products**
- 1.54 write ionic half-equations representing the reactions at the electrodes during electrolysis
- 1.55 recall that one faraday represents one mole of electrons**
- 1.56 calculate the amounts of the products of the electrolysis of molten salts and aqueous solutions.**